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SRX4700 100Gbps Full Duplex IPSEC tunnel

An HPE Juniper Networking Techpost

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1 SRX4700 100Gbps Full Duplex IPSEC tunnel

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The SRX4700 100Gbps Full Duplex IPSEC tunnel TechPost demonstrates the ability of the HPE Juniper Networking flagship 1RU firewall device to encrypt 100Gbps traffic patterns from a single system, such as a server or storage device, communicating within site-to-site tunnels. These single-tunnel, single-elephant flow use cases have never had practical solutions in the past: but this has now changed. Whilst this paper only concentrates on that use case (which was done for a customer performance Proof of Concept), it does not necessarily demonstrate the overall performance of the hardware-accelerated encryption enabled in Junos 25.2R1. The SRX4700 has significantly more headroom than what a single 100GE-connected system can practically produce. Typical use cases include backups, storage replication, and the transfer of large amounts of data between research organizations, whether supercomputer or AI-related workloads.



Figure 1: SRX4700 IPsec

1.1 Introduction

The SRX4700 is the next-generation 1RU fixed form factor firewall product that utilizes the Juniper Trio 6 ASIC (also known as YT silicon) as the Packet Forwarding Engine, along with two Intel Sapphire Rapids CPUs, combining resources for both the control and data planes. The Juniper YT ASIC is the same ASIC used in the MX range of routers and is widely deployed around the world. The flexibility of the YT ASIC with its unique fully programable forwarding path makes it an excellent accelerator for security use cases. These use cases range from large enterprise edge deployments to components of scale-out services for large service providers.

Two slices of the YT ASIC offer overall throughput of 1.6Tbps, on the SRX4700 divided between 1.4Tbps front panel Ethernet I/O and 200Gbps CPU connectivity. The idea is to process as much as possible within the

YT chip using Services Offload aka Express Path. This capability has been in SRX4600 and SRX5x00 for now more than a decade. What sets the YT ASIC apart is it now included crypto blocks that accelerate AES GCM encryption, and following on from MX terminology is called inline-IPSEC, documented here. The MX platforms, for instance the MX304, have been enabled with inline-IPSEC since Junos 24.4R1, delivering scalable IPSEC performance by utilizing multiple PFEs.

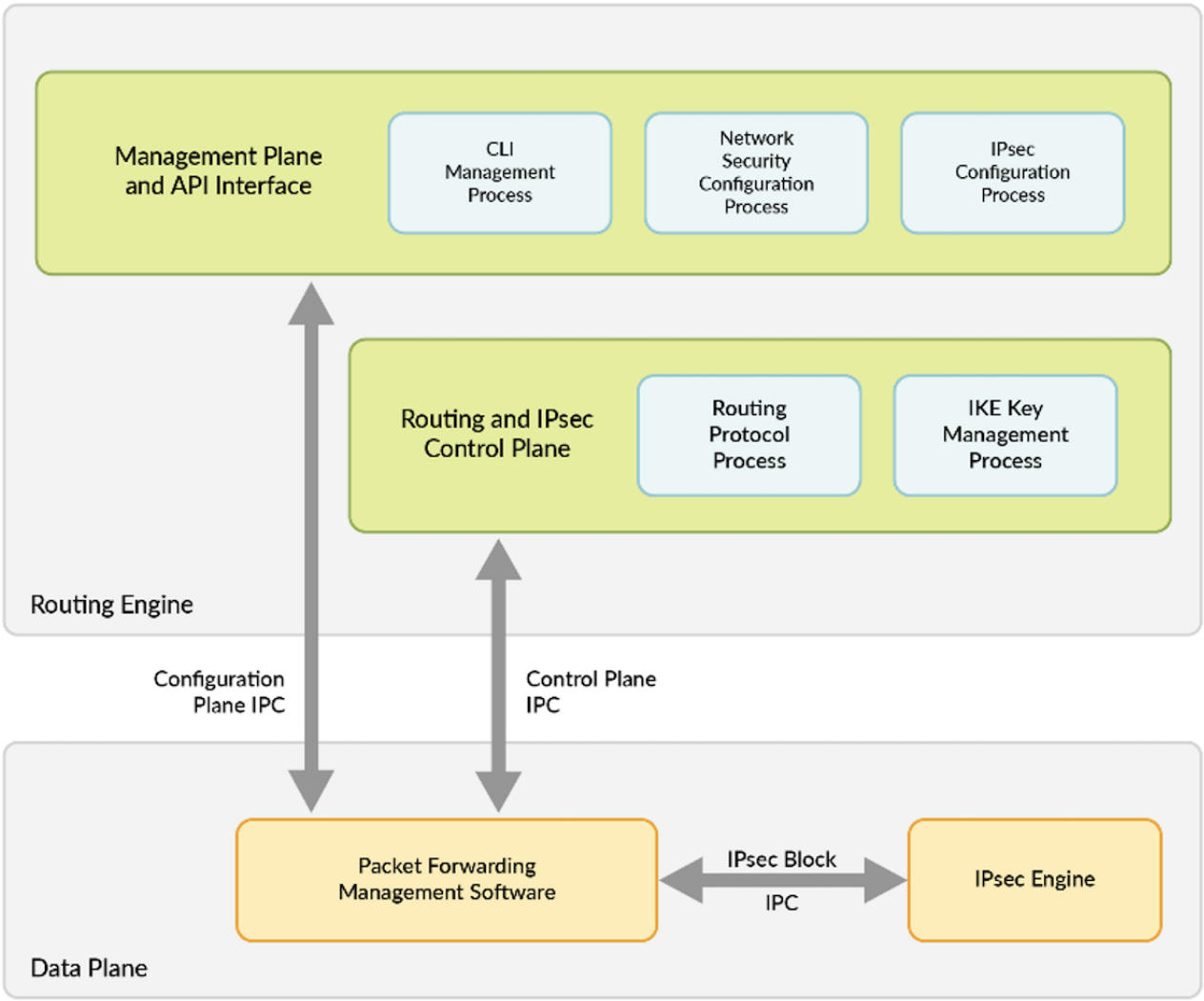


Figure 2: Generic block schema of inline-IPSEC architecture from Juniper documentation

Recently, there has been an increasing demand for high-throughput IPSEC tunnels, including elephant flows within those tunnels. As noted in the abstract, typical use cases include backup transfers to remote locations, distributed filesystems or storage replication, and the transfer of large amounts of data between research organizations, whether for supercomputer or AI-related workloads. The need for IPSEC arises in scenarios where MacSEC is not technically feasible (transport is L3 only) or policy deems the solution is insufficient (e.g., certification is required). Prior to the introduction of inline IPSEC, the technology used for high-speed tunnels on the SRX was Power Mode IPSEC, along with Symmetrical Fat Tunnel, often referred to as fat-core, which utilized multi-threaded processing of a single tunnel. With Junos 25.2R1, performance has increased by a clear order of magnitude due to processing in the YT ASIC, as demonstrated in the following text.

1.2 The Demo Setup

1.2.1 Topology

In a simplified manner, the demo consists of two back-to-back connected SRX4700 systems, each using a single 400GE interface with VLAN units for interconnection (ipsec zone) and the right and left zones towards a tester acting as the next hop for the respective /16 prefix. According to best practices, the route-based VPN tunnel interface is bound to its own zone (vpn). Hardware and software details are available in Appendix 2.



Figure 3: Logical topology schema

> Note: the physical topology, in addition to the SRX4700s, consists of QFX10k and PTX10001 devices for hosting 100GE and 400GE links that interconnect the devices and tester ports. The setup is not intended for latency measurements or for showcasing the full potential of the SRX4700's firewall and encryption capabilities.

1.2.2 SRX Configuration Break-down

The next section breaks down a simple, self-explanatory configuration file used for the demo case of a 100GE-connected tester utilizing an IPSEC tunnel between the two 400GE-connected SRX4700 devices. The entire configurations of both devices are present in Appendix 1.

1.2.3 SRX4700-203

Both PICs are configured in a mode with 1x400GE, 2x100GE, and 2x50GE (with 25GE, 10GE, 1GE capability):

```
set chassis fpc 1 pic 0 port-profile B-1X400G-2X100G-2X50G
set chassis fpc 1 pic 1 port-profile B-1X400G-2X100G-2X50G
```

IKE configuration classically defines encryption algorithms (with authentication in GCM modes embedded), DH (DHE) groups, lifetimes, pre-shared keys, and the remote peer, along with a bonus post-quantum security add-on in the form of RFC 8784 support, where this setup serves as a technology demonstrator with static keys:

```
set security ike proposal ike-1 authentication-method pre-shared-keys
set security ike proposal ike-1 dh-group group20
set security ike proposal ike-1 encryption-algorithm aes-256-gcm
set security ike proposal ike-1 lifetime-seconds 28800
set security ike policy ike-1 proposals ike-1
set security ike policy ike-1 pre-shared-key ascii-text "<SNIP>"
set security ike gateway 4700 ike-policy ike-1
set security ike gateway 4700 address 192.168.3.2
set security ike gateway 4700 external-interface et-1/1/0.1103
set security ike gateway 4700 version v2-only
set security ike gateway 4700 ppk-profile quantum
set security key-manager profiles quantum static key-id ascii-text "<SNIP>"
set security key-manager profiles quantum static key ascii-text "<SNIP>"
```

In the IPSEC part, the child Security Association (SA) configuration effectively, defines encryption algorithms, lifetimes, Perfect Forward Secrecy, adjusts anti-replay attributes, and binds previously defined IKE settings and the tunnel adapter. The extended sequence number prevents fast rollovers of sequence numbers, and high-speed tunnels typically require larger anti-replay windows, which are configured to the maximum supported. For the cipher set, AES GCM mode must be used to enable hardware offload:

```
set security ipsec proposal ipsec-1 extended-sequence-number
set security ipsec proposal ipsec-1 encryption-algorithm aes-256-gcm
set security ipsec proposal ipsec-1 lifetime-seconds 3600
set security ipsec policy ipsec-1 perfect-forward-secrecy keys group20
set security ipsec policy ipsec-1 proposals ipsec-1
set security ipsec vpn 4700 bind-interface st0.0
set security ipsec vpn 4700 ike gateway 4700
set security ipsec vpn 4700 ike anti-replay-window-size 4096
set security ipsec vpn 4700 ike ipsec-policy ipsec-1
set security ipsec vpn 4700 establish-tunnels immediately
```

Security policies and zone configuration are kept as simple as possible—permitting any traffic and minimal host-inbound services for establishing the VPN:

```
set security policies global policy permit match source-address any
set security policies global policy permit match destination-address any
set security policies global policy permit match application any
set security policies global policy permit then permit
set security zones security-zone ipsec interfaces et-1/1/0.1103 host-inbound-traffic system-
    services ping
set security zones security-zone ipsec interfaces et-1/1/0.1103 host-inbound-traffic system-
    services ike
set security zones security-zone left interfaces et-1/1/0.1101 host-inbound-traffic system-
    services ping
set security zones security-zone vpn interfaces st0.0
```

Finally, VLAN-tagged 400GE interface units are configured with static routing towards the remote (via unnumbered tunnel interface) and tester prefixes:

```
set interfaces et-1/1/0 vlan-tagging
set interfaces et-1/1/0 mtu 9000
set interfaces et-1/1/0 unit 1101 vlan-id 1101
set interfaces et-1/1/0 unit 1101 family inet address 192.168.1.1/24
set interfaces et-1/1/0 unit 1103 vlan-id 1103
set interfaces et-1/1/0 unit 1103 family inet address 192.168.3.1/24
set interfaces st0 unit 0 family inet
set routing-options static route 10.1.0.0/16 next-hop 192.168.1.10
set routing-options static route 10.2.0.0/16 next-hop st0.0
```

SRX4700-204

The VPN peer device is nearly identical, except for using the other 400GE interface (a matter of cabling; there's no difference), interface addressing, and reversed routing. As a side effect of using the alternative interface, the other crypto engine was utilized as well. For completeness, here's the diff of the SRX configurations, with the establish-tunnels immediately setting missing compared to the SRX4700-203:

“‘ diff SRX4700-203 SRX4700-204

10,11c10,11 < set security ike gateway 4700 address 192.168.3.2 < set security ike gateway 4700 external-interface et-1/1/0.1103 > set security policies global policy permit then permit > set security zones security-zone right interfaces et-1/0/0.1102 host-inbound-traffic system-services ping > set security zones security-zone ipsec interfaces et-1/0/0.1103 host-inbound-traffic system-services ping > set security zones security-zone ipsec interfaces et-1/0/0.1103 host-inbound-traffic system-services ike 34,42c33,41 < set interfaces et-1/1/0 vlan-tagging < set interfaces et-1/1/0 mtu 9000 < set interfaces et-1/1/0 unit 1101 vlan-id 1101 < set interfaces et-1/1/0 unit 1101 family inet address 192.168.1.1/24 < set interfaces et-1/1/0 unit 1103 vlan-id 1103 < set interfaces et-1/1/0 unit 1103 family inet address 192.168.3.1/24 < set interfaces st0 unit 0 family inet

< set routing-options static route 10.1.0.0/16 next-hop 192.168.1.10 < set routing-options static route 10.2.0.0/16 next-hop st0.0



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